# SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation Supplementary Material

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## A. Method and Training Details

### A.1. Error Decomposition

This subsection derives Eq. 2 and the rotation and context terms that Table S4 assigns to the tasks. It describes how the error of an estimate scales with the lead and the motion; it is not a model of success.

**Translation.** Write the true target position at the execution time as $\mathbf{p}(\tau^{\mathrm{a}}) = \mathbf{p}(\tau_t) + \mathbf{v}\Delta + \boldsymbol{\rho}$, with the velocity $\mathbf{v}$ at $\tau_t$ and the remaining motion $\boldsymbol{\rho}$, and the policy's estimate as $\hat{\mathbf{p}}(\tau^{\mathrm{a}}) = \hat{\mathbf{p}}(\tau_t) + \hat{\mathbf{v}}\hat{\Delta} + \hat{\boldsymbol{\rho}}$. Subtracting and adding $\mathbf{v}\hat{\Delta}$ gives

$$\begin{aligned}\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}}) = \big[\hat{\mathbf{p}}(\tau_t) - \mathbf{p}(\tau_t)\big] + (\hat{\mathbf{v}} - \mathbf{v})\hat{\Delta} \\ + \mathbf{v}(\hat{\Delta} - \Delta) + (\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}),\end{aligned} \tag{S1}$$

and the triangle inequality yields Eq. 2 with $e_v = \|\hat{\mathbf{v}} - \mathbf{v}\|$ and $e_{\mathrm{dyn}} = \|\hat{\boldsymbol{\rho}} - \boldsymbol{\rho}\|$. The bound holds for any split of the estimate into $\hat{\mathbf{v}}\hat{\Delta}$ and $\hat{\boldsymbol{\rho}}$, and $e_{\mathrm{dyn}}$ absorbs the higher-order part of the timing error. The residual $\boldsymbol{\rho}$ is the motion that constant-velocity extrapolation misses. Under a constant acceleration $\mathbf{a}$, $\boldsymbol{\rho} = \frac{1}{2}\mathbf{a}\Delta^2$, which an acceleration estimate from the history reduces; impacts, rolling contact, and friction changes leave a remainder that only a model of the dynamics can reduce.

**Rotation and the contact point.** Write the true orientation as $R(\tau^{\mathrm{a}}) = \mathrm{Exp}(\boldsymbol{\omega}\Delta + \boldsymbol{\eta})R(\tau_t)$, with the angular velocity $\boldsymbol{\omega}$ and the remaining rotation $\boldsymbol{\eta}$, and its estimate likewise. The rotation angle is bi-invariant, and Exp is 1-Lipschitz from $\mathbb{R}^3$ to SO(3), so for general rotations

$$\begin{aligned}\theta &= \big\| \mathrm{Log}(\hat{R}(\tau^{\mathrm{a}})R(\tau^{\mathrm{a}})^{\top})^{\vee} \big\| \\ &\le \theta_{\mathrm{loc}} + \|\hat{\boldsymbol{\omega}}\hat{\Delta} + \hat{\boldsymbol{\eta}} - \boldsymbol{\omega}\Delta - \boldsymbol{\eta}\| \\ &\le \theta_{\mathrm{loc}} + \hat{\Delta}\, e_\omega + \|\boldsymbol{\omega}\|\, e_\tau + e_\eta,\end{aligned} \tag{S2}$$

with the current orientation error $\theta_{\mathrm{loc}}$, $e_\omega = \|\hat{\boldsymbol{\omega}} - \boldsymbol{\omega}\|$, and $e_\eta = \|\hat{\boldsymbol{\eta}} - \boldsymbol{\eta}\|$. A contact point at $\mathbf{c}$ in the object frame, with $\|\mathbf{c}\| = r$, is off by $(\hat{\mathbf{p}} - \mathbf{p}) + (\hat{R}R^{\top} - I)R\mathbf{c}$ at $\tau^{\mathrm{a}}$, and the operator norm of $\hat{R}R^{\top} - I$ is $2\sin(\theta/2)$, so its error is

$$e_{\mathrm{c}} \le e + 2r\sin(\theta/2) \le e + r\theta, \tag{S3}$$

with $e = \|\hat{\mathbf{p}}(\tau^{\mathrm{a}}) - \mathbf{p}(\tau^{\mathrm{a}})\|$. A grasp or placement constrains several points spread over the body, such as the two finger contacts or the arms of a T-shaped part, so no single tracked point removes the term $r\theta$: the orientation error enters the contact scaled by the lever arm, and the target needs its full SE(3) state.

**Context.** Let $m$ be the largest margin that a feasible path can keep between the robot and the carried object and the context it passes, measured to the predicted positions of the context at $\tau^{\mathrm{a}}$. If the executed path keeps this margin, a collision requires a context error of at least $m$, so a coarse estimate of the surroundings suffices when $m \gg \varepsilon$. SE(3)-WAM does not enforce this margin, so the no-scene-flow control of Sec. 4.3 tests the effect of coarse context supervision; a flow sampled on a grid can also miss structures thinner than its spacing. A driver, such as a conveyor, a ramp, or a striker, acts on the contact through the motion it imparts to the target, which enters Eq. 2 through the target's own terms. The SE(3) history and forecast represent this motion precisely, so the driver needs no precise state of its own.

### A.2. Architecture

Table S1 lists the configuration of every module, in the order of the data flow described below.

**Scene tokens.** The frozen MAE ViT-B/16 [4] encodes the agent-view RGB into 196 patch tokens; we selected it on development scenes (Appendix E.1). Each patch token adds a view embedding and linear embeddings of its mean valid metric depth and of the base-frame position of its back-projected center pixel. These positions also drive 3D rotary embeddings of the patch queries and keys, which no other token receives, and the same center points are the source points of the sparse 3D flow. Past images never enter the model.

**State tokens.** The 8 target-state tokens sample the offsets $-350, -300, \ldots, 0$ ms. Each is a 25-dimensional descriptor $(\mathbf{p}, \log R, \Delta\mathbf{p}, \log \Delta R, \Delta t, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \boldsymbol{\alpha})$ with dimensions $(3, 3, 3, 3, 1, 3, 3, 3)$: slots 0–6 describe the seven intervals

Table S1. Architecture of SE(3)-WAM. Heads are skipped at deployment.

| Module | Configuration |
|---|---|
| Visual encoder | frozen MAE ViT-B/16, $224 \times 224$ input, 196 patch tokens |
| State tokenizer | $28 \to 1024 \to 1024$ GELU MLP, additive relative-time embedding |
| Tokens | 212 scene (196 patch, 8 target-state, 8 robot), 16 summary, 8 action |
| Backbone | 24 layers, width 1024, 16 heads, shared attention, SiLU FFN 4096 per stream, 504.0M parameters |
| Pose head | linear, mean summary state $\to 8 \times 6$ |
| Flow head | linear per patch token, $8 \times (3{+}1)$ |
| Appearance heads | linear per patch token: depth and moving-surface RGB as $16 \times 16$ pixels, 1024-d DINOv3 ViT-L/16 features |
| Target reconstruction | $1024 \to 256 \to 3072$ GELU MLP with sigmoid, $32 \times 32$ RGB |
| Action stream | flow matching, adaptive layer norm on the flow time, 5 Euler steps |

between consecutive offsets, with velocities from interval displacements over the elapsed time and accelerations from adjacent velocities over the spacing of the interval centers, and slot 7 holds the current absolute pose in the base frame. We standardize each dimension with training statistics, zero invalid dimensions, and append three validity flags (pose, interval, and acceleration); a slot without any valid feature is masked out of attention. On the real robot, we resample the tracker's estimates (Appendix D.2) at the same offsets, linearly in translation and along the geodesic in rotation, without extrapolation and without interpolating across gaps longer than two camera periods. The robot tokens hold the end-effector position, rotation vector, and measured gripper opening at the same offsets, and 16 learned embeddings form the summary tokens. The policy has no language token.

**Backbone and mask.** Each layer applies one attention over all tokens with query, key, value, and output projections shared by both streams, followed by separate SiLU feed-forward networks for the world and the action tokens. The mask of Sec. 3.3 gives a blocked edge the logit $-\infty$ in every layer and head, in training and at deployment, so the world states are computed once per query and reused across the Euler steps.

**Forecast and auxiliary heads.** All heads are readouts of the final world states, trained jointly with the action stream and skipped at deployment; no token reads their outputs. The pose head reads the mean summary state and is trained with $\mathcal{L}_P$ (Eq. 4). The flow head gives each patch point's displacement and visibility logit at every horizon, trained with $\mathcal{L}_F$ (Eq. 5) and the scale $s = 0.7154605\,\mathrm{m}$; the visibility loss has weight 1 in simulation, and points that are valid but invisible still supervise their displacement. The appearance heads predict at every horizon the metric depth with a squared error on valid pixels, the RGB of surfaces that move by more than 1 mm, including the robot, with a squared error, and the DINOv3 features [11] with a squared error after channel-wise layer normalization. The target-reconstruction head recovers the current target's RGB region from the mean summary state with an absolute error. These four losses form $\mathcal{L}_{\mathrm{aux}}$.

**Action stream.** Each of the 8 action tokens embeds a seven-dimensional command, a base-frame translation increment, a left-relative rotation vector, and a gripper command; the increments refer to the measured end-effector pose at application. At deployment, we integrate 5 Euler steps, reverse the normalization, decode the rotation vectors, and clip each axis to 0.05 m and 0.5 rad; the gripper follows the sign of its command. Appendix B.3 specifies how the chunks are executed.

### A.3. Training on Dynamic-14

Each control step of a demonstration anchors one window of 8 query positions with their own causal histories and no attention across positions, which gives 1,324,374 windows from the 11,200 demonstrations; 350 further episodes form the validation split. Command $j$ of the chunk for query $t$ is labeled with the demonstration command issued at $\tau_t + j\Delta_a$, since $d_{\mathrm{task}}$ applies only at execution (Eq. 1), and commands are normalized with the mean and standard deviation of the valid commands of the training split. Each query position draws 8 noise–time pairs for its chunk and averages their losses, which reduces the variance of the action gradient without enlarging the batch, and each loss is normalized by its count of valid labels in the global batch. The DINOv3 features of future frames are cached once per episode. Table S2 lists the recipe.

### A.4. Pretraining on DROID and Fine-Tuning

**Data.** The real-robot base trains on DROID [6], recorded at 15 Hz, with the 3D point tracks and camera calibrations that PointWorld released for it [5], matched to the raw episodes by exact agreement of robot states, commands, and camera identity. It uses the first exterior camera with its depth. Of 42,935 matched episodes, 42,087 yield 598,889 anchor windows, 538,740 from 37,874 training and 60,149 from 4,213 validation episodes. Each anchor expands into 8 consecutive query positions; the anchor carries the flow and DINOv3 labels, and the others carry action labels only. All times are capture timestamps, and the flow and DINOv3 targets keep PointWorld's record times. We keep the 96.88% of action

Table S2. Training hyperparameters on Dynamic-14.

| Item | Value |
|---|---|
| Optimizer | AdamW, $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay $10^{-4}$ |
| Learning rate | $2 \times 10^{-4}$, 5,174 warm-up updates from 0, cosine decay to 0 |
| Batch, updates | global batch 512 windows, 20 epochs (51,740 updates) |
| Loss weights | $\lambda_P = \lambda_F = 1$, $\lambda_R = 0.1$, $\lambda_{\text{aux}} = 0.1$ per target |
| Action noise | 8 noise–time pairs per query position |
| Precision | trainable weights FP32, frozen encoder BF16, gradient clipping at 0.1 |
| Inference | FP32, world states cached within a query |
| Hardware | [N] RTX PRO 6000 GPUs, [X] hours per run |

slots with an issued command, without a filter on speed, so that the base is not biased toward slow motion.

**Objectives.** DROID provides no object poses, so the base has no target-state tokens or pose head. It reads the current RGB-D frame and the robot history (204 scene tokens) and trains the action loss and the sparse 3D flow with weight 1 and the future DINOv3 features with weight 0.1; PointWorld records only visible positions, so the visibility loss is off. Most DROID objects move only when the robot moves them, so the base mainly learns metric geometry and the motion caused by the robot's own actions; the exogenous motion of our tasks and the target pose enter during fine-tuning.

**Flow labels.** Each patch of the $14 \times 14$ grid follows one PointWorld track: among the tracks that start inside the patch, are initially visible, and have a valid depth, the one with the largest image motion consistent with its 3D displacement, and otherwise the one nearest to the patch center, so that moving objects missed by a fixed center point are covered. The label is the track's displacement from its start; 47.05% of the point–horizon slots carry a label.

**Recipe and fine-tuning.** The base has 48 layers at width 1024 (1.02B trainable parameters) and trains for 20 epochs (21,060 updates, 2,106 warm-up updates) at a global batch of 512 on 64 GPUs, with the optimizer and precision of Table S2. Each real-robot policy starts from the base and is fine-tuned on the demonstrations of its task [updates, learning rate, and batch size]; the target-state tokens and the heads that the base lacks are freshly initialized. No real-robot policy uses simulation weights, and the base sees no simulation data.

# B. Dynamic-14 and Evaluation Protocol

## B.1. Tasks, Roles, and Success Conditions

We built Dynamic-14 on MuJoCo and robosuite [12, 18]. It contains fourteen tasks in five families (Fig. 4). Table S3 lists their success conditions and Table S5 their scenario ranges. We will release the tasks, the accepted scenarios, the success checks, and the demonstrations.

**Roles.** Table S4 assigns the bodies of each task to the target and context roles of Sec. 3.1. A moving receptacle is therefore a target, since its pose decides containment. The wire of the passage tasks plays both roles: its pose at the entry is a target, because the wand must be aligned within 30 mm there, and its geometry along the loop is context, because any contact fails. Conveyors, ramps, and strikers are drivers in the context. The last column lists the error terms of Eq. 2 that each task exercises (Appendix A.1). The two task groups of Fig. 5 follow from this table and were fixed before the ablations: the rotation-sensitive group contains the tasks whose terms include $r\theta$ (`grasp`, `t1_so3`, `t2_se3`, `t3_phase`, `t3_full`, `t4_can`, `t5_replan`), and the context-active group contains the tasks with an obstacle clearance or a driver (`t1_belt`, `t1_occ`, `t3_phase`, `t3_full`, and the four capture tasks and `t5_replan`).

**Terminal events.** We check success at every 2 ms physics step, after the ordered requirements of Table S3, and an interrupted continuous condition resets its counter. Terminal events fall into five categories. *Invalid* events are corrupted simulator states. *Boundary* events are a workspace or downstream exit, a drop, or an expired goal window. *Safety* events are unsafe contact, an excessive rim impulse, wire contact, a release impulse, early robot contact, unsupported tray contact, or a striker trap. *Task violations* are a release outside the goal, a path deviation or backtracking, a premature release, a deflection without capture, tipping, a missing or extra impact, or an insufficient change of outgoing motion. The last category is the *timeout*. Each family evaluates its events in a fixed order per physics step and keeps the first terminal event.

## B.2. Scenarios, Demonstrations, and Splits

We sample continuous ranges uniformly and discrete choices with equal probability, and then apply physical-feasibility checks. Evaluation replays the stored accepted scenarios. Harmonic translation follows $\mathbf{p}(t) = \mathbf{c} + (0, v_y, 0)t + (A_x \sin(2\pi f_x t + \phi), 0, 0)$ with $\phi \sim \mathcal{U}[-\pi, \pi]$. Capture targets start at rest and move under gravity and friction on ramps of $[5, 15]^\circ$. We accept an occluded scene only if the target is visible for at least eight initial frames, fully occluded for 0.5–3.0 s, and visible again for at least three frames.

Table S3. Success conditions of the fourteen Dynamic-14 tasks, with the thresholds of our task implementations. A task must pass its preceding stages and avoid terminal failures. All tasks run at 20 Hz with a 2 ms physics step; durations below refer to physical time. COM denotes center of mass.

| Task | Ordered success conditions | Limit |
|---|---|---|
| grasp; t1_belt, t1_xyz, t1_so3, t1_occ | Approach within 0.10 m; bilateral fingerpad contact for 0.05 s; collision-geometry bottom at least 0.05 m above support; retain contact and clearance for 0.50 s. T1 adds conveyor drive, translation, rotation, or a physical occluder. SO3 has no additional success-angle threshold; unsafe occluder contact terminates OCC. | 160 steps 8 s |
| t2_trans | Approach within 0.10 m. Entire object collision geometry inside the receptacle: planar clearance and floor gap both $\geq -0.002$ m, top no higher than the opening. Release requires an opening command $< -0.5$ and no fingerpad contact. Relative linear/angular speeds $\leq 0.08$ m/s and 0.8 rad/s for 0.10 s, followed by a 0.50 s stable dwell. | 200 10 s |
| t2_se3 | Approach within 0.05 m and 0.12 rad. Full T-shaped containment: planar clearance $\geq 0.002$ m, floor gap in $[-0.002, 0.002]$ m, tilt $\leq 0.05$ rad, top below opening. After release, relative speeds $\leq 0.02$ m/s and 0.04 rad/s for 0.10 s, then a 0.50 s dwell. The 0.12 rad gate applies to approach. | 200 10 s |
| t3_phase | The episode starts with the wand already grasped. Phase alignment: distance $\leq 0.090$ m and loop-direction error $\leq 0.20$ rad. Enter within 0.030 m at progress $\leq 0.15$; reach historical maximum progress $\geq 0.60$, then $\geq 0.85$ and endpoint distance $\leq 0.055$ m. | 420 21 s |
| t3_full | Acquire the wand with bilateral contact for 0.05 s, then complete the phase stages above. In a later sample, retain current progress $\geq 0.85$ and move at least 0.045 m from both path and endpoint. Finally open ($< -0.5$) with no fingerpad contact. Releasing before exit fails. | 640 32 s |
| t4_sphere | Approach within 0.10 m; first qualifying contact after ramp exit; any tray contact before success fails. Bilateral hold for nominal 25 ms (12 physics samples, 24 ms). Raise COM by 0.05 m from its height when the bilateral-hold criterion is first met, and retain contact and height for 0.50 s. | 120 6 s |
| t4_sphere_tabletop; t4_slider | Approach within 0.10 m; contact after ramp exit, while supported, in the capture region, and moving at planar speed $\geq 0.08$ m/s. Bilateral hold for 0.05 s, COM lift $\geq 0.05$ m from the first attainment of that hold criterion, then contact and height for 0.50 s. | 120 6 s |
| t4_can | Same supported-capture sequence. Before first fingerpad contact, a can-axis absolute dot product with vertical $> 0.45$ fails as unstable tipping. The configuration does not require a transverse closing axis. | 120 6 s |
| t5_replan | Exactly one selected-side impact group completes before robot contact. At 30 ms after the physical impact-end timestamp, branch-signed lateral speed $\geq 0.035$ m/s and yaw rate $\geq 0.35$ rad/s. Approach within 0.10 m; bilateral hold with grasp-axis error $\leq 18^\circ$ for 0.05 s; COM lift $\geq 0.05$ m from the first attainment of that hold criterion and stable hold for 0.50 s. Impact-group lag is 12 ms and maximum contact is 0.10 s. | 120 6 s |

The training pool contains 11,200 demonstrations, 800 per task, generated by experts with privileged simulator state; the learned policies receive only the inputs of Sec. 3. We record the demonstrations at 20 Hz with camera images, depth, robot states, issued commands, and simulator ground truth for target poses and scene geometry, from which we compute the future labels, their validity masks, and the command labels.

We select configurations on 1,400 development scenes (100 per task). All Dynamic-14 results use a separate held-out set of 1,400 scenes generated with the same ranges and new source seeds that appear in no training, validation, or development episode; the first 100 feasible scenes in increasing seed order form each task's set.

### B.3. Execution Protocol and Latency Measurement

**Executor.** Every method runs under the executor of Sec. 3.4 on a 50 ms control clock, with queries starting on its period boundaries. The total latency $L_t$ of query $t$ is the time that physics advances while it runs: its inference time under the protocol plus any added delay. Command $j$ of its chunk executes at $\tau^{\mathrm{a}}_{t,j}$ of Eq. 1, with $d_{\mathrm{task}} = 50$ ms for the post-impact grasp and zero otherwise. Its commands $j < k_t$, with $k_t = \lceil L_t/\Delta_a \rceil$, have expired and are discarded, so query $t$ executes commands $j = k_t, \ldots, \min(8, k_t + k_{t+1}) - 1$, one per control period. When $k_t + k_{t+1} > 8$, e.g., for a constant total latency above 200 ms, a chunk runs out before the next one arrives, and if $k_t \geq 8$, all its commands have expired. Before the first chunk of an episode arrives, and whenever a chunk runs out, the controller holds the current arm target and gripper command while physics advances;

Table S4. **Roles and tolerances of the Dynamic-14 tasks.** The target is the body whose pose error enters a contact or containment tolerance $\varepsilon$. The context comprises obstacles, avoided with clearance $m$, and drivers that move the target. Speeds follow the sampling ranges of Table S5 (capture tasks: at the ramp exit). Terms: error terms that a task exercises (Eq. 2 and Appendix A.1). $v$: moving target. $r\theta$: orientation enters the contact. $e_{\mathrm{dyn}}$: non-constant motion (harmonic, gravity, friction, impact). $m$: a context body must be avoided. Occ.: history gaps. Delay: command delay built into the task.

| Task | Code | Target | Context | $\varepsilon$ (mm) | $\|\mathbf{v}\|$ (m/s) | $m$ (mm) | Terms |
|---|---|---|---|---|---|---|---|
| Harmonic grasp | grasp | block | table | [X] | 0.03–0.20 | – | $v$, $r\theta$, $e_{\mathrm{dyn}}$ |
| Conveyor grasp | t1_belt | part | belt | [X] | 0.04–0.10 | – | $v$ |
| Translating grasp | t1_xyz | part | table | [X] | 0.04–0.15 | – | $v$, $e_{\mathrm{dyn}}$ |
| Rotating grasp | t1_so3 | part ($\omega \leq 1.2$ rad/s) | table | [X] | 0.04–0.15 | – | $v$, $r\theta$, $e_{\mathrm{dyn}}$ |
| Occluded grasp | t1_occ | part | occluder canopy | [X] | 0.04–0.10 | [X] | $v$, $m$, occ. |
| Moving receptacle | t2_trans | receptacle | table | [X] | 0.03–0.06 | – | $v$ |
| Rotating receptacle | t2_se3 | T-shaped receptacle | table | [X] | 0.03–0.06 | – | $v$, $r\theta$ |
| Wire passage | t3_phase | wire loop (entry) | rotating wire | 30 | [X] | swept | $v$, $r\theta$, $m$ |
| Wire passage, full | t3_full | wire loop (entry) | rotating wire | 30 | [X] | swept | $v$, $r\theta$, $m$ |
| Ball capture | t4_sphere | ball | tray, ramp | [X] | [0.4–0.7] | [X] | $v$, $e_{\mathrm{dyn}}$, $m$ |
| Ball capture, table | t4_sphere_tabletop | ball | table, ramp | [X] | [0.4–0.7] | – | $v$, $e_{\mathrm{dyn}}$ |
| Slider capture | t4_slider | slider | table, ramp | [X] | [0.4–0.8] | – | $v$, $e_{\mathrm{dyn}}$ |
| Can capture | t4_can | can | table, ramp | [X] | [0.4–0.7] | – | $v$, $r\theta$, $e_{\mathrm{dyn}}$ |
| Post-impact grasp | t5_replan | part | striker | [X] | 0.12–0.14 | [X] | $v$, $r\theta$, $e_{\mathrm{dyn}}$, $m$, delay |

no command is repeated. The reported time-to-action is the measured latency before rounding to $k_t\Delta_a$.

**Real-time protocol.** Unless noted otherwise, physics advances during inference by the method's native latency, drawn per query from its latency distribution measured at batch size 1 on one RTX 5090. The native-latency ACT/DP rollouts run on RTX 5090; ACT paused and added-latency rollouts and the planned remaining evaluations use RTX PRO 6000. Their own compute time does not enter the episode. Table S8 reports three alternatives. With *paused* physics, the executor is unchanged and the simulation waits for inference; an added delay still advances physics. Without added delay, $L_t = 0$, and command 0 executes at capture. With *equal latency*, every method is charged the p95 latency of SE(3)-WAM regardless of its own speed. With *original executors*, each baseline keeps the execution scheme of its paper, e.g., a longer chunk before replanning.

**Added latency.** An added latency of 0, 50, 100, 150, or 200 ms is inserted after inference on top of the protocol's latency, and physics advances during it.

**Latency measurement.** We time all methods on one RTX 5090 (32 GB), the GPU of our robot, with exclusive use and batch size 1. These measurements form the latency distributions that the simulated rollouts replay independently of the rollout hardware. For each method, each of three rounds uses thirty warm-up calls and 300 measured calls over three observation sequences of eight frames, with CUDA synchronization at call boundaries and a rotated method order. Each method uses the precision of its release; SE(3)-WAM uses FP32. The time-to-action includes encoding the image and generating the first command, and on the robot also tracking the newest frame. For SE(3)-WAM, we time the forecast heads separately, although deployment skips them. We distinguish three times. The *time-to-action* is the measured wall-clock time from the arrival of an observation to the availability of its first command, without rounding; the *handover time* $\ell_t = k_t\Delta_a$ rounds the total latency up to the control period; and the *action age* is the execution time of a command minus the capture time. On the robot, traces record capture, availability, preprocessing, the start and end of inference, queue entry, command issue, and execution. A query misses its deadline when its time-to-action exceeds the 50 ms control period, so that the next period executes a remaining command or holds. Appendix C.2 reports the measured values.

### B.4. Baselines and Ablation Controls

**Common conditions.** Table S6 summarizes the adaptations. Each baseline keeps its official architecture, action chunk, and initialization. ACT and Diffusion Policy train from scratch, $\pi_{0.5}$ is fine-tuned from `pi05_base`, and the other methods start from their released weights. All of them train on the same 11,200 demonstrations, whose commands we map to each action space through the same Cartesian controller. The seven trained baseline endpoints use agent-view RGB only, as does SE(3)-WAM. The native missing-view paths are retained: $\pi_{0.5}$ masks its two zero wrist slots; DreamVLA uses a preprocessed black wrist image with wrist auxiliary supervision masked; OpenWAM-$\alpha$ and DynamicWAM fill their missing wrist regions with black. Each baseline follows its official update budget, trained once with seed 0; the input and action adaptations are stated here and in

Table S5. Scenario proposal ranges. Length, linear velocity, angle, and angular velocity use m, m/s, rad, and rad/s unless marked otherwise. Positions are in world coordinates except the stated ramp-local offsets. Continuous intervals are uniform proposals; discrete choices are equiprobable before feasibility checks. Evaluation replays the stored accepted scenarios, whose distribution need not remain uniform. Harmonic centers are not initial positions.

| Task | Initial-state proposal | Motion or event parameters |
|---|---|---|
| grasp | Harmonic center $c_x \in [-.08, .08]$, $c_y \in [-.26, -.19]$, $z = .822$; yaw $[-1.1, 1.1]$. | $v_y \in [.025, .125]$, $A_x \in [.015, .035]$, $f_x \in [.3, .7]$ Hz; $\omega_z \in [-1.2, 1.2]$. |
| t1_belt | Target $x \in [-.08, .08]$, $y \in [-.26, -.18]$, $z = .822$; identity orientation, zero initial twist. | Belt $v_y \in [.04, .10]$. |
| t1_occ | Same target proposal as Belt; canopy at $(.322, -.29, 1.20)$. | Same belt range; clearance parameter .20; physical occlusion. |
| t1_xyz | Harmonic center $c_x \in [-.025, .025]$, $c_y \in [-.24, -.20]$, $z = .818$; yaw $[-.35, .35]$. | $v_y \in [.04, .09]$, $A_x \in [0, .03]$, $f_x \in [.25, .65]$ Hz; $\omega_z = 0$. |
| t1_so3 | Same as XYZ. | Same translation; $\omega_z \in [-1.2, -.4] \cup [.4, 1.2]$, equal sign probabilities. |
| t2_trans | Goal $x \in [-.13, -.08]$, $y \in [-.18, -.12]$, $z = .825$; fixed orientation. | Goal $v_y \in [.03, .06]$; no rotation. |
| t2_se3 | Same goal $x, y$ range, $z = .804$; yaw $[-.3, .3]$. | Goal $v_y \in [.03, .06]$; $\omega_z \in [-.25, -.08] \cup [.08, .25]$, equal sign probabilities. |
| t3_phase | Wand $x, y \in [-.03, .03]$, $z = .870$, yaw $[0, 2\pi]$; wire phase $[-.25, .35]$. | Wire center $(.10, -.02, .94)$; world-X rotation $\omega_x = -\{2.5, 3, \ldots, 7.5\}^\circ$/s (11 equal-probability values). |
| t3_full | Wand at $(-.05, -.02, 1.04)$ with identity orientation; same wire phase range. | Same wire rotation; cradle and release requirement enabled. |
| t4_sphere | Ramp-local offsets: lateral $[-.015, .015]$, longitudinal/normal $(-.145, .030)$; lip $(.06, -.02, 1.10)$. | Gravity-driven from rest; gate release at tick 0; slope $[5, 15]^\circ$; target/ramp sliding friction $[.7, .9]$; unsupported capture. |
| t4_sphere_tabletop | Lateral $[-.015, .015]$, longitudinal/normal $(-.124, .034)$; lip $(0, -.06, .808)$. | Same slope and release; target/ramp friction $[.7, .9]$, table friction .35. |
| t4_slider | Lateral $[-.015, .015]$, longitudinal/normal $(-.129, .024)$; supported ramp. | Same slope and release; target/ramp friction $[.025, .045]$, table friction $[.012, .018]$. |
| t4_can | Lateral $[-.010, .010]$, ramp-axis yaw $[-4, 4]^\circ$; longitudinal position follows gate geometry and oriented can extent. | Same slope and release; target/ramp friction $[.25, .45]$, table friction $[.09, .15]$. |
| t5_replan | $x \in [-.005, .005]$, $y = -.29$, $z = .812$; yaw $[-.08, .08]$; zero initial twist. | Belt $v_y \in [.12, .14]$; equal-probability left/right striker triggered at $y = -.13$; stroke .098, pulse .35 s, actuator scale $[1, 1.1]$, force cap 4 N; command delay 50 ms. |

Table S6. For DreamVLA, we ran both its original 40-epoch schedule and a 20-epoch cosine schedule and report the one that was better on development scenes.

**Inputs.** PUMA [2] receives its historical optical flow, computed on the rendered frames. DreamVLA receives seven causal RGB frames, the end-effector position and orientation, the finger positions, and a fixed task text, and it samples with ten DDIM steps and classifier-free guidance of 1.5. OpenWAM-$\alpha$ receives the current RGB views, joint values, gripper opening, and task text, and it generates video and actions jointly with video decoding disabled. DynamicWAM receives its flow history and kinematic descriptors.

**SE(3)-augmented baselines.** $\pi_{0.5}$+SE(3) and DynamicWAM+SE(3) receive the eight target-state tokens of Appendix A.2 from the same pose source, with the same validity flags and masking. A two-layer MLP maps the tokens to the model's width. For $\pi_{0.5}$, we append them to the prefix after the image tokens, where the action expert attends to them; for DynamicWAM, they join its kinematic descriptors. The output layer of the MLP is zero-initialized, and training follows the base model's recipe.

**Ablation controls.** The five controls of Sec. 4.3 change one component each of the configuration of Table S2 and keep the data, the recipe, the update budget, the executor, and the training seeds 0, 1, and 2. *No target history* removes slots 0–6 of the target-state tokens, which carry the past poses

Table S6. **Baseline adaptations.** All methods train on the same 11,200 demonstrations with the inputs listed below and run under the shared executor of Appendix B.3. Updates × batch follows the official recipe. Exec.: the execution scheme of the original paper (commands executed per chunk before replanning), used only for the original-executor control of Table S8. Params: trainable parameters, frozen encoders excluded. Seeds: training seeds.

| Method | Initialization | Inputs | Updates × batch | Exec. | Params | Seeds |
|---|---|---|---|---|---|---|
| ACT [17] | scratch | current agent RGB, robot state | 50,000 × 32 | temporal agg. | [X] | 1 |
| Diffusion Policy [1] | scratch | two causal agent RGB frames, robot state | 50,000 × 32 | 8 | [X] | 1 |
| $\pi_{0.5}$ [9] | `pi05_base` | current agent RGB, robot state, task text; wrist slots masked | 30,000 × 256 | 10 | [X] | 1 |
| $\pi_{0.5}$ + SE(3) | `pi05_base` | planned: $\pi_{0.5}$ + SE(3) tokens; camera inputs pending | 30,000 × 256 | 10 | [X] | 1 |
| DynamicVLA [15] | released DOM ckpt. | frames $[-2, 0]$, robot state, task text | 50,000 × 44 | streaming | [X] | 1 |
| PUMA [2] | official checkpoint | RGB, optical-flow history, robot state, task text | [X]×[X] | [X] | [X] | 1 |
| DreamVLA [16] | released components | seven causal agent RGB frames, robot state, task text; black wrist input | 48,220 × 512 | 1 | [X] | 1 |
| OpenWAM-$\alpha$ [13] | official foundation | current agent RGB, joints, gripper, task text; missing views black | 34,490 × 192 | 32 | [X] | 1 |
| DynamicWAM [8] | official initialization | RGB, flow history, kinematic descriptors | 40,000 × 128 | 16 | [X] | 1 |
| DynamicWAM + SE(3) | official initialization | planned: DynamicWAM + SE(3) tokens; camera inputs pending | 40,000 × 128 | 16 | [X] | 1 |
| SE(3)-WAM | frozen MAE encoder | current agent RGB-D, SE(3) and robot histories | 51,740 × 512 | 1 | 523.8M | 3 |

with their velocity and acceleration features, and keeps slot 7 with the current pose (Appendix A.2). *No future supervision* removes $\mathcal{L}_P$, $\mathcal{L}_F$, and the three future appearance losses with their heads, and keeps the reconstruction of the current target region. *No scene-flow supervision* removes $\mathcal{L}_F$ and the flow readout, and *no dedicated SE(3)* removes $\mathcal{L}_P$ and the pose head; both keep the target-state tokens and all appearance targets. The *uniform-fidelity forecast* replaces $\mathcal{L}_P$ and $\mathcal{L}_F$ by [a forecast of future latents of a frozen video VAE of the whole frame at the same horizons, its readout, and its loss weight]. Since every forecast head is skipped at deployment, only the no-history control changes the inputs at inference.

### B.5. Metrics and Statistics

The task-balanced success rate is

$$\mathrm{SR}(m) = \frac{100}{14} \sum_{k=1}^{14} \frac{1}{S_m n_k} \sum_{s=1}^{S_m} \sum_{i=1}^{n_k} y_{mski}, \qquad \text{(S4)}$$

for binary outcomes $y_{mski}$ of method $m$, trained with seed $s$, on scene $i$ of task $k$, with $S_m = 3$ training seeds for our variants and $S_m = 1$ for the baselines. The *position error at first contact* $e_p$ is the distance of the end effector from the expert's contact frame on the target at the first contact. The two placement tasks start with the object held, so their contact is the first contact of the held object with the receptacle, and their contact frame is the end-effector pose implied by the expert's placement target. The phase-only passage (`t3_phase`) succeeds without any contact, so $e_p$ averages over the other 13 tasks; we report it on episodes that make contact, and failures without contact stay in the SR denominator.

**Training seeds and intervals.** The full model and the five ablation controls use training seeds 0, 1, and 2, and each baseline uses seed 0. On development scenes, three training seeds of the same configuration differed by up to [7.4] points in SR, mostly on the wire passage, whose success ranged from [0] to [71]% across seeds. We report each training seed and the cross-seed mean and standard deviation (Appendix C.4); for the ablations, we also repeat the comparison without the phase-only wire passage (`t3_phase`). For paired 95% intervals, we average the seeds per scene and bootstrap held-out scenes within each task, jointly across methods. These intervals quantify scene uncertainty for the trained weights, and we treat differences whose interval lies within $\pm 2$ points as ties.

## C. Additional Simulation Results

### C.1. Per-Task Results

Table S7 breaks the held-out Dynamic-14 results of Table 1 down by task for every method, with the paired interval of SE(3)-WAM against the strongest other method on each task; its last two rows give the position error at first contact $e_p$ and the number of episodes with a contact (Appendix B.5). With its native 100-step DDPM sampler, every chunk of Diffusion Policy has expired when it arrives, so none of its commands is executed. [Tasks on which every method fails or a baseline outperforms SE(3)-WAM, and the dominant failure of SE(3)-WAM.]

### C.2. Evaluation Protocols and Latency

Table S8 evaluates every method under three further protocols. Paused physics shows how the relative performance changes when the cost of computation is ignored, equal

Table S7. **Per-task held-out success** (%) on Dynamic-14 for every method of Table 1, mean of three training seeds for our variants and one for baselines, real-time protocol. DP: Diffusion Policy. DVLA: DynamicVLA. DrVLA: DreamVLA. OWAM: OpenWAM-$\alpha$. DWAM: DynamicWAM. +: with our SE(3) history tokens. $\sigma$: SE(3)-WAM with nominal synthetic correlated pose noise (Appendix C.5). $\Delta$: SE(3)-WAM minus the strongest other method on that task, with its 95% paired interval. The last two rows give the position error at first contact $e_p$ (mm), averaged over the 13 tasks with a contact, and the number $n$ of episodes with a contact (Appendix B.5); –: no 13-task mean. Task codes are listed in Table S4.

| Task | ACT | DP | $\pi_{0.5}$ | $\pi_{0.5}$+ | DVLA | PUMA | DrVLA | OWAM | DWAM | DWAM+ | $\sigma$ | **Ours** | $\Delta$ [95% CI] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Harmonic grasp | 48.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Conveyor grasp | 46.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Translating grasp | 18.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Rotating grasp | 21.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Occluded grasp | 32.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Moving receptacle | 26.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Rotating receptacle | 0.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Wire passage | 99.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Wire passage, full | 0.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Ball capture | 36.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Ball capture, table | 71.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Slider capture | 57.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Can capture | 90.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Post-impact grasp | 44.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| Task-balanced | 42.0 | 0.0 | — | — | — | — | — | — | — | — | — | — | — |
| $e_p$ (mm) | 32.2 | – | — | — | — | — | — | — | — | — | — | — | – |
| $n$ | 1109 | 6 | — | — | — | — | — | — | — | — | — | — | – |

latency compares all methods at the p95 latency of SE(3)-WAM, and the original executors check whether the shared executor removes a capability that a baseline relies on. With paused physics, the difference between SE(3)-WAM and [name] is [±X.X] points, and under equal latency it is [±X.X] points. The original executors change the SR of the baselines by at most [X.X] points. Table S9 gives the success at every added latency; its rows for SE(3)-WAM and the strongest baseline are the data of Fig. 1 (right).

Table S8. **Protocol controls** on held-out Dynamic-14 scenes (SR, %). Paused: shared executor, with physics stopped during inference. Equal latency: shared executor, with every method charged the p95 latency of SE(3)-WAM. Original executor: real time with each method's own execution scheme. Real time: the protocol of Table 1. Late: queries in real time whose time-to-action exceeds the 50 ms control period (%). Age: p95 action age in real time (ms); –: no command executed.

| Method | Paused | Equal latency | Original executor | Real time | Late (%) | Age p95 |
|---|---|---|---|---|---|---|
| ACT | 40.9 | — | — | 42.0 | 0.0 | 50 |
| Diffusion Policy | — | — | — | 0.0 | 100.0 | – |
| $\pi_{0.5}$ | — | — | — | — | — | — |
| $\pi_{0.5}$ + SE(3) | — | — | — | — | — | — |
| DynamicVLA | — | — | — | — | — | — |
| PUMA | — | — | — | — | — | — |
| DreamVLA | — | — | — | — | — | — |
| OpenWAM-$\alpha$ | — | — | — | — | — | — |
| DynamicWAM | — | — | — | — | — | — |
| DynamicWAM + SE(3) | — | — | — | — | — | — |
| SE(3)-WAM | — | — | – | — | — | — |

Table S9. **Success versus added latency** on held-out Dynamic-14 scenes (SR in %, real-time protocol). The 0 and 150 ms columns equal the Native and +150 ms columns of Table 1.

| | Added latency (ms) | | | | |
|---|---|---|---|---|---|
| Method | 0 | 50 | 100 | 150 | 200 |
| ACT | 42.0 | — | — | 24.1 | — |
| Diffusion Policy | 0.0 | — | — | 0.0 | — |
| $\pi_{0.5}$ | — | — | — | — | — |
| $\pi_{0.5}$ + SE(3) | — | — | — | — | — |
| DynamicVLA | — | — | — | — | — |
| PUMA | — | — | — | — | — |
| DreamVLA | — | — | — | — | — |
| OpenWAM-$\alpha$ | — | — | — | — | — |
| DynamicWAM | — | — | — | — | — |
| DynamicWAM + SE(3) | — | — | — | — | — |
| SE(3)-WAM | — | — | — | — | — |

The native-latency ACT/DP rollouts ran on RTX 5090; ACT paused and ACT/DP added-latency rollouts ran on RTX PRO 6000. Each method reuses its recorded RTX 5090 latency distribution.

**Timing.** Table S10 breaks the time-to-action of SE(3)-WAM on one RTX 5090 down by stage, measured as in Appendix B.3; Table 1 gives the p95 latency of every method and Table S6 the parameter counts. On the robot, we report the p50, p95, and maximum action age and the fraction of all queries of the test trials that miss the 50 ms deadline, together with the peak memory.

Table S10. **Latency breakdown** of one query of SE(3)-WAM on one RTX 5090 at batch size 1 (ms, p50 / p95), and action age on the robot.

| Stage | p50 | p95 |
|---|---|---|
| Pose tracking (per frame, robot) | — | — |
| Image encoding | — | — |
| World stream (once per query) | — | — |
| Action stream (5 Euler steps) | — | — |
| Time-to-action (total) | — | — |
| Forecast heads (training only, for reference) | — | — |
| Action age at execution (robot) | — | — |
| Queries over 50 ms (robot, %) | — | |

## C.3. Results on DOMINO

On DOMINO [2], we compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of its clean level-1 setting. All methods use the benchmark's demonstrations, scenes, and native evaluator, one training seed, and 50 episodes per task and method. We report SR and the manipulation score (MS), which scales route completion by penalties for leaving the workspace or view and for collisions (Table 2). SE(3)-WAM uses domain-specific normalization and reads the simulator pose of the target named by the instruction; slots without a valid pose are masked as in Appendix A.2. The RGB-based baselines do not receive this pose. [Summarize the DOMINO results.]

## C.4. Final Ablations and Seed Stability

Table S11 reports the five controls of Sec. 4.3 (Appendix B.4) for each training seed, with the cross-seed mean and standard deviation and the paired difference from the full model, with native latency and with 150 ms of added latency. The last two columns repeat the paired differences without the phase-only wire passage (`t3_phase`), whose success varied most across training seeds (Appendix B.5). The two task groups of Fig. 5 are listed in Appendix B.1. [Whether any conclusion of Sec. 4.3 changes for an individual seed or without `t3_phase`.]

## C.5. Robustness to Pose Noise

With frozen weights, we evaluate three synthetic perturbations of the target pose: nominal temporally correlated noise, twice its amplitude, and nominal noise with contiguous missing observations. The nominal process combines a fixed per-episode bias and a stationary correlated component, each with per-axis standard deviations of 5 mm and 1°, with a correlation time of 200 ms. Dropouts last 200 ms, separated by geometrically distributed valid gaps with a mean of 3.8 s, giving a long-run missing fraction of 5%. The perturbation applies to the current and past estimates before the history features are computed.

Fig. S1 evaluates the final model under these perturbations, on the same held-out scenes at zero added latency and with all three training seeds. Relative to clean poses, SR changes by [X.X], [X.X], and [X.X] points under nominal correlated noise, twice its amplitude, and missing observations.

**Figure placeholder: Fig. S1 — pose noise**
x: clean poses, nominal correlated noise, twice its amplitude, and nominal noise with missing observations; y: SR of SE(3)-WAM per task family, frozen weights, real-time protocol; paired intervals over matched scenes.

Figure S1. **Robustness to pose noise.** SR of SE(3)-WAM under synthetic perturbations of the target pose, relative to clean poses.

## C.6. Forecast Diagnostics

Table S12 compares the learned SE(3) forecast of the final model offline with constant-velocity and constant-acceleration extrapolation from the same pose history.

Table S11. **Final ablations by training seed** on held-out Dynamic-14 scenes (Appendix C.4). S0–S2: SR (%) of training seeds 0–2; mean ± s.d. over seeds; Δ: control minus SE(3)-WAM (points) with its paired 95% interval. Last two columns: Δ without the phase-only wire passage (`t3_phase`), 13 tasks.

| | Native | | | | | +150 ms | | | | | Δ w/o `t3_phase` | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Variant | S0 | S1 | S2 | Mean ± s.d. | Δ [95%] | S0 | S1 | S2 | Mean ± s.d. | Δ [95%] | Native | +150 ms |
| SE(3)-WAM * | — | — | — | — | – | — | — | — | — | – | – | – |
| No target history | — | — | — | — | — | — | — | — | — | — | — | — |
| No future supervision | — | — | — | — | — | — | — | — | — | — | — | — |
| No scene-flow supervision | — | — | — | — | — | — | — | — | — | — | — | — |
| No dedicated SE(3) | — | — | — | — | — | — | — | — | — | — | — | — |
| Uniform-fidelity forecast | — | — | — | — | — | — | — | — | — | — | — | — |

Table S12. **Offline prediction errors.** Target-position error at 400 ms (mm) on the same held-out observations. CV and CA are analytic extrapolations from causal pose history; the learned head reads frozen full-model states.

| | Const. vel. | Harmonic | Roll/slide | Impact |
|---|---|---|---|---|
| CV extrapolation | — | — | — | — |
| CA extrapolation | — | — | — | — |
| Learned pose head | — | — | — | — |

# D. Real-Robot Experiments

## D.1. Setup and Tasks

The robot is a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera, commanded through [a timestamped Cartesian interface] at [control frequency]. The camera records RGB-D at [resolution] and [frame rate], and we calibrate its extrinsics by [procedure] before each session. SE(3)-WAM uses this fixed third-person view; baselines whose recipes require a wrist view receive [per-method camera configuration]. All policies run on one RTX 5090. Fig. S2 shows the layout, and Table S13 defines the six tasks. T1 and T2 share a belt conveyor, T3 uses a turntable, T4 a remotely driven toy car, T5 a cue-driven collision between two balls, and T6 a ramp. Touching or pushing the target without completing the task, a single-finger hold, and an action after the target has left the workspace count as failures; [equipment-invalid trials and safety interventions are repeated or counted as failures by the following rule].

## D.2. Pose Tracking and Demonstration Labels

Tracking is marker-free. Grounding DINO [7] localizes the object named by the task in the first frame, SAM 2.1 [10] gives its initial mask, and FoundationPose [14] initializes the object's 6D pose from this mask and then tracks it. [FoundationPose variant and mode (model-based with an object mesh or model-free with reference views), how the object frame is kept consistent across demonstrations and trials, and re-initialization after a loss.] The tracker runs on the control clock: at every control-period boundary, it updates the pose on the newest camera frame captured at or before the boundary, so the pose history has the 50 ms spacing used in training, and each estimate keeps the capture time of its frame. A query waits for the update of its own boundary and then runs the policy on the same GPU. While a query spans further boundaries, the tracker keeps updating at a higher GPU priority and processes only the newest frame, so that an update never queues behind older ones; the tracking time and this contention are part of the measured latency (Table S10). [Camera frame rate and the measured time per update.] When the target is lost, the estimate is marked invalid, and the previous pose is not carried forward. [How the target poses of the demonstrations, used as inputs and training labels, are obtained.] For the ball tasks, the target-state tokens keep the translation, velocity, and acceleration of the ball center and mask its orientation and angular motion, and the pose head is supervised on translation only. Table S14 reports the tracking accuracy.

## D.3. Training and Trial Protocol

We compare SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and Diffusion Policy [1], with 20 trials per task and method. All methods use the same pre-generated trial keys, which fix the object instance, the initial pose, and the motion parameters, sampled by [sampling protocol]. Trials run in [randomized method order], so that drift of the apparatus does not align with the methods. An operator resets the scene. A trial starts at the first usable observation, the robot may move at any time, and a trial ends at success, failure, or a timeout of [X] s. Demonstrations, development trials, and test trials come from separate sessions. SE(3)-WAM is fine-tuned per task from the DROID base (Appendix A.4), and the baselines are trained or fine-tuned on the same demonstrations with [per-method initialization and training settings]. For every method, we log the end-to-end time-to-action of each query on the same RTX 5090, including tracking for SE(3)-WAM; Table 4 reports its p95.

**Figure placeholder: Fig. S2 — real-robot setup**

**(a) Left, photo of the workcell** from behind the robot: the third-person RGB-D camera, the robot, and the conveyor, turntable, car track, ball-collision track, and ramp at their operating positions, each labeled; the wrist camera if a baseline uses one. **(b) Right, top-down schematic to scale:** the robot base frame with its axes, the camera position and viewing direction with its distance to the workspace, the footprint and motion direction of each device for T1–T6, the grasp or interception region of each task, and the workspace bounds in cm. **Rules:** real photo, unretouched; schematic from the calibrated extrinsics and measured device positions; no repetition of the task photos of Fig. 6.

Figure S2. **Real-robot setup.** (a) The workcell. (b) Top-down layout of the camera, the robot base frame, and the devices of T1–T6 [with the workspace dimensions].

Table S13. Real-robot task definitions. $N = 100$: training demonstrations per task; 20 additional demonstrations per task are reserved for validation.

| Task | Device, motion, and objects | Operating range; $N$ | Success and failure criteria |
|---|---|---|---|
| T1 | Conveyor transports a graspable cube. | [belt speed]; 100 | [stable grasp and lift; failures] |
| T2 | Conveyor transports an open box; the arm places a cube into it. | [belt speed]; 100 | [placement and retention; failures] |
| T3 | Turntable carries a cube around its axis. | [angular speed, radius]; 100 | [stable grasp and lift; failures] |
| T4 | Remotely driven toy car moves on the tabletop. | [speed, path]; 100 | [car grasp and lift; failures] |
| T5 | A cue-driven ball strikes a second ball; the arm intercepts the second ball after the collision. | [impact and outgoing speed]; 100 | [post-impact capture; failures] |
| T6 | A ball rolls or a block slides down a ramp toward the interception region. | [slope, release, speed]; 100 | [capture and retention; failures] |

Table S14. **Real-robot pose tracking.** Median/p95 translation (mm) and rotation (°) error of the tracker against a marker-free reference, the uncertainty of that reference, and the fraction of lost observations. Ball rotation is not used.

| Target | mm | ° | Ref. uncertainty | Lost |
|---|---|---|---|---|
| Real rigid object | — | — | — | — |
| Real ball | — | – | — | — |

## D.4. Detailed Results and Failures

Table S15 reports successes per task, 95% binomial intervals for the pooled rates, and paired differences from a bootstrap over trial keys. [The most frequent failures of SE(3)-WAM and their causes.]

Table S15. **Real-robot successes** per task (successes/trials, 20 trials per task; tasks in Table S13). All: pooled success rate (%) with its 95% binomial interval. Δ: paired difference from SE(3)-WAM (points) with a bootstrap interval over trial keys.

| Method | T1 | T2 | T3 | T4 | T5 | T6 | All | Δ |
|---|---|---|---|---|---|---|---|---|
| $\pi_{0.5}$ | — | — | — | — | — | — | — | — |
| DynamicWAM | — | — | — | — | — | — | — | — |
| Diffusion Policy | — | — | — | — | — | — | — | — |
| SE(3)-WAM | — | — | — | — | — | — | — | – |

# E. Preliminary Design Studies

We screened the encoder, the history representation, the forecast targets, and the coupling of world and action streams with smaller models: 4 layers of width 768, trained for 4 epochs (40,104 updates at a global batch size of 16) and evaluated with the synchronous executor that preceded our real-time protocol. Each group keeps its own encoder and seed count, stated below, and conditions are compared within a group. These studies selected the components of the final model; the five controls of Sec. 4.3 then test its central components at full scale and under the real-time protocol (Appendix C.4).

## E.1. Encoder, Backbone, and History

**Encoder and backbone.** We compared a CNN trained with the policy, frozen DINOv3 ViT-S/16 and ViT-L/16 [11], and MAE ViT-B/16 [4] (three seeds for the MAE reference; Table S16), and selected MAE. We also compared a backbone with shared attention and stream-specific feed-forward networks against one with fully separate world and action parameters, at about 47.3M backbone parameters and with three seeds each ([source of the backbone comparison]), and selected the shared attention.

**History.** With a frozen DINO-L encoder and one seed per condition, we compared the current state alone, the target-motion history, RGB windows, Farnebäck flow [3], flow with motion statistics [8], codec motion vectors, video-VAE latents, and compressed windows (Table S17). Its last row replaces the past target poses of the trained history model by its current pose at test time.

## E.2. Forecast Targets and Auxiliary Supervision

With a frozen DINO-L encoder and one seed per condition, we compared fifteen future targets, from target-only and image-space forecasts to role-aware combinations (Table S18). A second group with the MAE encoder and one seed removed each future appearance target from the selected configuration. In the first group, no future target improved clearly on the variant without future supervision (P00, 62.78% against 62.38% for scene flow and SE(3)); single seeds at this scale do not resolve such differences, and the full-scale ablations of Sec. 4.3 test future supervision in the final model.

Table S16. **Observation encoders** at the screening scale (Appendix E.1). Params: encoder parameters; p95: time-to-action.

| Encoder | Params | Trained | SR | $SR_{150}$ | p95 |
|---|---|---|---|---|---|
| CNN | — | yes | — | — | — |
| DINOv3 ViT-S/16 | 22M | frozen | — | — | — |
| DINOv3 ViT-L/16 | 300M | frozen | — | — | — |
| MAE ViT-B/16* | 86M | frozen | — | — | — |

Table S17. **History representations** at the screening scale (frozen DINO-L encoder, one training seed, 1,400 scenes). The first eight conditions receive the current target pose; the ninth adds the target-motion history. Last row: the ninth model with its past target poses replaced by the current pose at test time and the kinematic features recomputed.

| History input | Success / 1,400 | SR (%) |
|---|---|---|
| Current image | 748 | 53.43 |
| RGB window | 662 | 47.29 |
| Optical flow | 720 | 51.43 |
| Flow + motion statistics | 723 | 51.64 |
| Codec motion vectors | 733 | 52.36 |
| Video-VAE latent | 710 | 50.71 |
| Compressed window (age) | 634 | 45.29 |
| Compressed window (uniform) | 670 | 47.86 |
| Current image + target-motion history | 824 | 58.86 |
| stationary history at test time | 9 | 0.64 |

## E.3. World–Action Coupling

With the MAE encoder and three seeds per condition, we compared six couplings of the world and action streams. R0 disables every forecast loss. R1 adds one object query and one query per patch to the world stream, which attend to $\mathcal{S}$, $\mathcal{M}$, and each other and carry the forecast readouts, without any edge to the actions. R2 lets the action tokens attend to these queries at every layer, R3 lets the queries attend to the actions, and R4 opens both edges; in R3 and R4, the world states are recomputed at every Euler step. R5 decodes the forecast once per query and encodes the target trajectory and the flow with visibility, with rotations in a continuous 6D form, through a geometry encoder (0.37M parameters) into tokens that the actions attend to. Future supervision without

Table S18. **Forecast targets** at the screening scale. The first group uses a frozen DINO-L encoder and seed 0, with its 140 development scenes and the 1,260 remaining scenes reported separately. The second group uses the MAE encoder and seed 1 on 1,400 scenes; it keeps scene flow, SE(3), and the current-target RGB reconstruction and removes future appearance targets. Compare conditions within each group. Targets are cumulative from the current state at the eight horizons, except codec motion: SE(3), the target pose; 3D scene flow, base-frame displacements of the $14 \times 14$ patch centers; 2D tracks and dense 2D displacement, image displacements of target points, grid points, or all pixels; codec motion, the motion vectors of a video codec; video-VAE latent, latents of a frozen Wan2.2 VAE; object DINO, mask-pooled DINOv3 features of the target.

| ID | Future supervision | Dev. success / 140 | Success / 1,260 | $SR_{1260}$ (%) |
|---|---|---|---|---|
| P00 | None | 88 | 791 | 62.78 |
| P01 | 3D scene flow | 95 | 782 | 62.06 |
| P02 | SE(3) | 92 | 789 | 62.62 |
| P03 | Scene flow + SE(3) | 93 | 786 | 62.38 |
| P04 | Video-VAE latent + SE(3) | 92 | 783 | 62.14 |
| P05 | Dense 2D displacement + SE(3) | 87 | 746 | 59.21 |
| P06 | Codec motion + SE(3) | 86 | 795 | 63.10 |
| P07 | Sparse scene 2D tracks + SE(3) | 90 | 807 | 64.05 |
| P08 | Scene flow + object DINO | 89 | 794 | 63.02 |
| P09 | Scene flow + target 2D tracks | 88 | 808 | 64.13 |
| $A_\alpha$ | P03 + motion magnitude | 89 | 778 | 61.75 |
| P12 | P03 + depth | 90 | 791 | 62.78 |
| P13 | P03 + DINO | 92 | 788 | 62.54 |
| P14 | P03 + dynamic RGB | 89 | 790 | 62.70 |
| P15 | P03 + depth / DINO / dynamic RGB | 94 | 802 | 63.65 |
| AUX0 MAE-B: future appearance auxiliaries | | Success / 1,400 | | $SR_{1400}$ (%) |
| None | | 894 | | 63.86 |
| Depth only | | 867 | | 61.93 |
| DINO only | | 946 | | 67.57 |
| Dynamic RGB only | | 926 | | 66.14 |
| All three (existing reference) | | 995 | | 71.07 |

Table S19. **Couplings R0–R5** at the screening scale (MAE encoder, three training seeds each, 1,400 development scenes per seed; Appendix E.3). SR: mean over seeds (%); s.d.: standard deviation over seeds (points).

| Coupling | SR | s.d. |
|---|---|---|
| R0 (future losses off) | 60.45 | 3.57 |
| R1 (queries, no edge) | 63.60 | 2.34 |
| R2 (world→action) | 62.52 | 1.97 |
| R3 (action→world) | 64.21 | 2.13 |
| R4 (bidirectional) | 63.00 | 4.19 |
| R5 (decoded geometry) | 63.95 | 1.91 |

any edge to the actions (R1) exceeded R0 by 3.14 points on average, with seed-wise differences of −1.29, +9.64, and +1.07. Feeding the forecasts to the actions gave no consistent further gain: R2 and R5 differed from R1 by −1.07 and +0.36 points on average, with standard deviations of 4.28 and 3.45 points over the seed-wise differences, and the effects of both cross-stream edges changed sign across seeds (Table S19). We therefore use the forecasts only as training supervision, which also keeps deployment free of their cost.

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